

Parity-Covariant Gluing and Fixed-Depth Obstructions for Native Möbius Rows

A Hostile Reconstruction of a Two-Row Mellin–Landau Route

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17 August 2026

Abstract

A projective paired-source proposal attempted to prove coefficientwise nonnegativity of a canonical Möbius row by stopping a rough-prime recursion at one-prime leaves, applying one canonical Target–Lorenz transport to every leaf, and then invoking a two-row Mellin–Landau argument. We give a standalone hostile reconstruction of the complete chain.

The coefficient magnitudes and every component-row activation indicator are transported exactly through arbitrary rough-prime histories. The missing interface is a parity cocycle: a history of length m multiplies the signed terminal marginal by $(-1)^m$. The published terminal realization is one-sided and is not invariant under this cocycle. We prove a necessary and sufficient parity-covariant gluing theorem and exhibit the explicit odd-history leaf

$$X = 67 \cdot 71 \cdot 13, \quad h = (67), \quad (p, y) = (71, 13).$$

Exact dyadic rational interval arithmetic over all 239 active P_{61} divisors proves

$$E_T(71, 13) - O_T(71, 13) > 17,$$

whereas the odd history requires the reverse Target–Lorenz orientation.

We next show that the obvious parity repair also fails. Iterating any fixed depth L gives an exact disjoint source decomposition, but its current block has leading sign $(-1)^{L-1}(\log \log X)^{L-1}$. Hence an even depth restores canonical recursive parity only by making the current block eventually negative. A 256-bit MPFR certificate already gives

$$B_{2,200000}(2) < -11, \quad B_{2,200000}(3) < -2.$$

Thus no fixed-depth local reset can close the route.

As a clean replacement we prove a global parity-aware common-source gluing theorem. At each finite endpoint it either produces one source-faithful positive row after all history parities are incorporated, or emits a separating dual certificate. Its two-row specialization GPHT_{2,3} is the exact remaining global producer. Finally, we reprove the fixed-row reciprocal-zeta Mellin transform, a self-contained Landau lemma, and the fact that rows two and three have no common open-strip cancellation. Therefore GPHT_{2,3} implies the Riemann Hypothesis. The producer is not proved here. The unconditional content of the paper is the corrected gluing theorem, the explicit counterexample, the fixed-depth no-go theorem, the global primal-dual architecture, and the exact Mellin–Landau reduction. The Riemann Hypothesis remains unproved.

Scientific status. This manuscript is an unconditional correction and reduction, not an announcement of a proof of the Riemann Hypothesis. It withdraws the parity-blind composition used in the predecessor full-row proposal and isolates a new parity-aware producer.

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1 Introduction

The route studied here starts from a family of positive component rows $Q_X(j)$ and forms the signed Möbius row

$$c_X(j) = \sum_{k \leq X/j} \frac{\mu(k)}{\sqrt{k}} Q_{X/k}(j), \quad j \geq 2. \quad (1)$$

A proposed proof of $c_X(j) \geq 0$ for all X and j used three ideas:

- (i) encode the sign $\mu(k)$ as the difference of two positive parity channels;
- (ii) expand squarefree k by ordered rough-prime histories and stop when the remaining quotient lies below 67;

- (iii) realize every stopped one-prime leaf by the same positive Target–Lorenz map, then sum the resulting positive rows.

The downstream analytic implication is unusually short. If rows two and three are nonnegative, their Mellin transforms contain $1/\zeta(s+1/2)$, while their finite Mellin numerators cannot vanish simultaneously at a point of the open critical strip. A nonnegative-transform theorem of Landau then excludes every zero with real part greater than $1/2$.

The burden is therefore entirely upstream: does the positive source tree really produce the signed row in (1) without changing signs, copying source atoms, or dropping activations? This paper reconstructs that interface at atom level.

The answer separates cleanly into two parts. Magnitudes and activations are correct. If a rough prime p is moved from the source index into the path, then

$$p^{-1/2}(k/p)^{-1/2} = k^{-1/2}, \quad \frac{X/p}{k/p} = \frac{X}{k}.$$

The sign is not invariant: the parity channels are exchanged once, and the signed observation changes sign. After m rough primes, the terminal datum is $(-1)^m$ times its canonical local orientation. The predecessor proof retained this parity in its labels but omitted it from the terminal realization.

Our main unconditional contributions are:

- a necessary-and-sufficient parity-covariant projective gluing theorem;
- an exact proof that rough placement preserves every row activation and coefficient magnitude while contributing the cocycle $(-1)^{|h|}$;
- a proof-grade odd-history obstruction with target gap greater than 17;
- an exact depth- L source expansion and a no-go theorem for every fixed local parity reset;
- a global parity-aware common-source primal-dual theorem;
- a self-contained two-row Mellin–Landau consumer.

The corrected architecture does not terminalize one-prime leaves independently and does not attempt another fixed-depth reset. It expands the finite source at a fixed endpoint, incorporates every cumulative history parity, and applies one global common-source transport. The resulting producer is GPHT_{2,3}. Separate leafwise certificates cannot prove it, because they would reuse source capacity and erase the parity cocycle.

2 Canonical component rows and native Möbius rows

2.1 The component row

For $j \geq 2$, set

$$A_j = \frac{j+1}{j-1}, \quad B_j = \frac{(j+1)(j-2)}{j(j-1)}, \quad C_j = \frac{2}{j(j-1)}.$$

The canonical component row is

$$\begin{aligned} Q_X(j) &= \frac{A_j}{\sqrt{j}} \log \frac{X}{j} \mathbf{1}_{X \geq j} - \frac{B_j}{\sqrt{j+1}} \log \frac{X}{j+1} \mathbf{1}_{X \geq j+1} \\ &\quad + C_j \sum_{m \geq j+2} \frac{1}{\sqrt{m}} \log \frac{X}{m} \mathbf{1}_{X \geq m}. \end{aligned} \tag{2}$$

Every sum in this paper is causally extended by zero. In particular, $Q_X(j) = 0$ when $X < j$. Between consecutive integer knots, $Q_X(j)$ is affine in $\log X$.

The full native row is (1). No sign assertion is built into its definition. The exact ordinary-capacity and entropy identities that motivated the row are useful background, but the present paper uses only the component formula (2) and its Mellin transform.

2.2 Squarefree source notation

Let

$$P_{61} = \prod_{p \leq 61} p.$$

Every squarefree source index has a unique factorization

$$k = d p_1 \cdots p_t, \quad d \mid P_{61}, \quad 67 \leq p_1 < \cdots < p_t. \quad (3)$$

We call $h(k) = (p_1, \dots, p_t)$ its ordered rough history. Since k is squarefree,

$$\mu(k) = \mu(d)(-1)^t. \quad (4)$$

The factor $(-1)^t$ is the cocycle omitted by the parity-blind gluing step.

3 Positive paired sources and the parity cocycle

3.1 Paired source category

A paired source is a pair $P = (E, O)$ of positive measures. Let

$$S(E, O) = (O, E)$$

be the parity swap. Given a positive row map R , define signed observation by

$$\mathcal{O}(P) = R(E) - R(O). \quad (5)$$

The fundamental identity is immediate.

Lemma 3.1 (Parity cocycle). *For every integer $m \geq 0$,*

$$\mathcal{O}(S^m P) = (-1)^m \mathcal{O}(P). \quad (6)$$

Proof. The claim for $m = 1$ follows by exchanging the two terms in (5); induction gives the result. $\square$

3.2 Activation and coefficient covariance

The concern that arbitrary rough histories might move a row across an activation boundary is natural. It is not the failing interface.

Theorem 3.2 (Exact rough-placement functor). *Suppose $p \mid k$ is one rough factor. Moving p from the source index into the path sends (X, k) to $(X/p, k/p)$ and multiplies the child coefficient by $p^{-1/2}$. Then*

$$\frac{X/p}{k/p} = \frac{X}{k}, \quad p^{-1/2}(k/p)^{-1/2} = k^{-1/2}. \quad (7)$$

Consequently every indicator $\mathbf{1}_{X/k \geq j}$ and every value $k^{-1/2}Q_{X/k}(j)$ is transported without change in magnitude. After a history h the signed value is

$$(-1)^{|h|} k^{-1/2} Q_{X/k}(j). \quad (8)$$

Proof. Both identities in (7) are algebraic. The row activation depends only on the ratio X/k . The sign statement follows from (6). $\square$

Remark 3.3. *The theorem covers one-sided activation boundaries exactly. If $X/k = j$, then both parent and placed child lie on the same boundary. No continuity or floating-point argument is involved.*

4 The corrected projective gluing theorem

We now state the abstract interface in the form needed for a rigorous source composition.

Definition 4.1 (Labelled projective source tree). *A labelled projective source tree consists of:*

- (a) *a finite or finitely terminating rooted tree;*
- (b) *at each node v , a positive paired source P_v and an ordered rough history $h(v)$;*
- (c) *a one-use direct-sum source identity expressing each nonterminal P_v as current source plus positive multiples of child sources;*
- (d) *at every terminal node, a canonical local signed datum K_v .*

The root-oriented terminal datum is

$$\varepsilon(v)K_v, \quad \varepsilon(v) = (-1)^{|h(v)|}.$$

Theorem 4.2 (Parity-covariant projective gluing). *Assume every source atom has one owner and every recursive coefficient is nonnegative. A family of positive terminal rows preserves the exact root marginal if each terminal row realizes*

$$\varepsilon(v)K_v. \tag{9}$$

Conversely, replacing a terminal datum by a positive row realizing K_v while $\varepsilon(v) = -1$ does not preserve the root marginal unless $K_v = 0$ in every observed coordinate.

Proof. Apply signed observation to the direct-sum identity at every node. By (6), the observation of the terminal source at v is $\varepsilon(v)K_v$. Substitution of exact terminal realizations therefore preserves the root identity. Since all coefficients are nonnegative and ownership is one-use, the substituted sum is positive.

For necessity, suppose $\varepsilon(v) = -1$ but the installed row realizes K_v . Choose an observed coordinate on which $K_v \neq 0$. The correct contribution is $-K_v$ and the installed contribution is K_v , changing the root coordinate by $2K_v$. No label bookkeeping can remove this change after the terminal source has been replaced. $\square$

Corollary 4.3 (When parity-blind gluing is legal). *A terminal map proved only for canonical orientation can be applied to every leaf only if all stopped histories are even, or if both K_v and $-K_v$ have separately proved positive realizations.*

Firewall 4.4. *Storing the parity in the source label is not sufficient. The terminal realization itself must use the label through (9).*

5 An explicit odd-history obstruction

5.1 The terminal target atoms

The canonical one-prime terminal construction uses the target kernel

$$T(u) = (4\sqrt{u} - 3)\mathbf{1}_{u \geq 1}. \quad (10)$$

For a terminal prime $p \geq 67$, child endpoint $1 \leq y < 67$, and small divisor $d \mid P_{61}$, define

$$t_d(p, y) = \frac{1}{\sqrt{d}} \left[T(py/d) - \frac{1}{\sqrt{p}} T(y/d) \right]. \quad (11)$$

The canonical even and odd target masses are

$$E_T(p, y) = \sum_{\substack{d \mid P_{61} \\ \mu(d)=1}} t_d(p, y), \quad O_T(p, y) = \sum_{\substack{d \mid P_{61} \\ \mu(d)=-1}} t_d(p, y). \quad (12)$$

Inactive terms vanish causally.

The Target–Lorenz map used in the predecessor route is oriented from canonical even supply to canonical odd demand. It selects an even submeasure of target mass O_T and leaves a positive even residual. Thus it requires $E_T \geq O_T$.

5.2 The path $67 \rightarrow 71$ at endpoint 61841

Take

$$X = 67 \cdot 71 \cdot 13 = 61841. \quad (13)$$

After the first rough factor 67, the node endpoint is $923 = 71 \cdot 13$. The next terminal current leaf has $(p, y) = (71, 13)$. The incoming history has length one, so Theorem 4.2 reverses the canonical orientation. Actual even supply is the canonical odd mass $O_T(71, 13)$, while actual odd demand is the canonical even mass $E_T(71, 13)$.

Theorem 5.1 (Directed target obstruction). *For $(p, y) = (71, 13)$,*

$$E_T(71, 13) - O_T(71, 13) > 17. \quad (14)$$

Consequently the reversed exact-target Hall condition $O_T(71, 13) \geq E_T(71, 13)$ is impossible.

Proof. There are 239 active squarefree divisors $d \mid P_{61}$ with $d \leq 923$. The supplied verifier encloses every square root by exact dyadic rationals with denominator 2^{192} . For a rational $a/b > 0$, it computes

$$n = \left\lfloor \sqrt{\lfloor a2^{384}/b \rfloor} \right\rfloor$$

and uses

$$\frac{n}{2^{192}} \leq \sqrt{a/b} \leq \frac{n+1}{2^{192}},$$

with equality detected by exact integer arithmetic. Interval addition, multiplication, subtraction, and reciprocal are then rational operations. Applying these operations to (11) and summing by Möbius parity yields

$$\begin{aligned} E_T(71, 13) &\in [238.236769968802241851178078781701 \dots, \\ &\quad 238.236769968802241851178078781702 \dots], \\ O_T(71, 13) &\in [221.231683430583189260479623028094 \dots, \\ &\quad 221.231683430583189260479623028095 \dots]. \end{aligned}$$

The resulting rational lower bound for the difference exceeds 17. The complete rational endpoints are retained in the deterministic verification object; the decimals above are display strings only. $\square$

Corollary 5.2 (Failure of parity-blind terminalization). *The canonical Target–Lorenz terminal map cannot be substituted on the leaf (13) after history (67) while preserving the complete target datum. Therefore the parity-blind projective composition does not prove $c_X(j) \geq 0$.*

Remark 5.3. *The conclusion is deliberately narrow. It refutes the proof interface, not the inequality $c_X(j) \geq 0$. Different rough histories may cancel globally. Any successful successor must retain that global cancellation or provide a new positive joint source decomposition.*

6 Fixed-depth parity resets and their obstruction

The odd-history obstruction suggests advancing the source by an even number of rough primes before recursing. Source-theoretically this is exact. Positivity, however, imposes a second and independent constraint.

Let $p_j, p_{j+1}, \dots$ be the allowed ordered rough primes. Let $P_j(x)$ be the corresponding paired source, and let $b_j(x)$ be the base consisting of histories with no allowed rough prime. Unique least-prime factorization gives

$$P_j(x) = b_j(x) \oplus \bigoplus_{k \geq j} p_k^{-1/2} S P_{k+1}(x/p_k). \quad (15)$$

Theorem 6.1 (Exact depth- L expansion). *For every fixed integer $L \geq 1$,*

$$\begin{aligned} P_j(x) = & \bigoplus_{r=0}^{L-1} \bigoplus_{j \leq k_1 < \dots < k_r} (p_{k_1} \dots p_{k_r})^{-1/2} S^r b_{k_r+1} \left(\frac{x}{p_{k_1} \dots p_{k_r}} \right) \\ & \oplus \bigoplus_{j \leq k_1 < \dots < k_L} (p_{k_1} \dots p_{k_L})^{-1/2} S^L P_{k_L+1} \left(\frac{x}{p_{k_1} \dots p_{k_L}} \right), \end{aligned} \quad (16)$$

where the $r = 0$ term is $b_j(x)$. The partition is disjoint and exhaustive. The recursive packets have parity $(-1)^L$ and endpoint at most $x/(p_j \dots p_{j+L-1})$.

Proof. Classify a finite rough history by whether its length is smaller than L . A history of length $r < L$ has unique ordered primes $p_{k_1} < \dots < p_{k_r}$ and terminates in the corresponding base. A history of length at least L has a unique first L -tuple and a remaining history in P_{k_L+1} . Coefficients factor multiplicatively and the swaps contribute S^r or S^L . $\square$

For $P = P_{61}$, write the positive finite-Euler row

$$D_{P,Y}(q) = \sum_{d|P} \frac{\mu(d)}{\sqrt{d}} Q_{Y/d}(q). \quad (17)$$

After signed observation, the current block in (16) at rough threshold 67 is

$$B_{L,X}(q) = \sum_{\substack{m \leq X/q \\ m \text{ squarefree} \\ p|m \Rightarrow p \geq 67 \\ \omega(m) < L}} \frac{\mu(m)}{\sqrt{m}} D_{P,X/m}(q). \quad (18)$$

For $L = 2$ this is the root base minus all one-rough-prime bases.

Lemma 6.2 (Finite-Euler principal term). *For each fixed row $q \geq 2$,*

$$D_{P,Y}(q) = a_q \sqrt{Y} + O_q(\log(2Y)), \quad a_q = \frac{8}{q(q-1)} \prod_{\substack{\ell \leq 61 \\ \ell \text{ prime}}} \left(1 - \frac{1}{\ell}\right) > 0. \quad (19)$$

Proof. The tail in (2) is a Riemann sum for

$$\int_1^Y t^{-1/2} \log(Y/t) dt = 4\sqrt{Y} - 4 - 2\log Y.$$

Integral comparison for the positive decreasing summand gives

$$Q_Y(q) = 4C_q \sqrt{Y} + O_q(\log(2Y)).$$

Insert this in the finite sum (17). The principal coefficient is

$$4C_q \sum_{d|P} \frac{\mu(d)}{d} = \frac{8}{q(q-1)} \prod_{\ell|P} (1 - \ell^{-1}).$$

□

We use the following standard fixed-order form of the almost-prime estimates. It is included to make the obstruction independent of numerical evidence.

Lemma 6.3 (Reciprocal rough almost-primes). *Fix $r \geq 1$ and a fixed prime threshold z . Put*

$$H_z(X) = \sum_{z \leq p \leq X} \frac{1}{p}.$$

For squarefree integers all of whose prime factors are at least z ,

$$\sum_{\substack{m \leq X \\ \omega(m)=r}} \frac{1}{m} = \frac{H_z(X)^r}{r!} + O_{r,z}(H_z(X)^{r-1}), \quad (20)$$

$$\sum_{\substack{m \leq X \\ \omega(m)=r}} \frac{\log(2X/m)}{\sqrt{m}} = O_{r,z} \left(\frac{\sqrt{X} H_z(X)^{r-1}}{\log X} \right). \quad (21)$$

Moreover $H_z(X) = \log \log X + O_z(1)$.

Proof. Mertens' theorem for primes gives the last assertion. Expanding the r -fold prime harmonic sum, deleting repeated primes, imposing increasing order, and then imposing the product cutoff changes the result by $O_{r,z}(H_z^{r-1})$; this proves (20) by induction on r .

For (21), the same induction gives the classical fixed-order Landau bound

$$A_r(t) := \#\{m \leq t : \omega(m) = r\} \ll_{r,z} \frac{t(\log \log t)^{r-1}}{\log t}.$$

Partial summation against the decreasing function $t^{-1/2} \log(2X/t)$, splitting at $\sqrt{X}$, yields the displayed bound. All constants may depend on the fixed r and z . □

Theorem 6.4 (Fixed-depth parity-reset no-go). *For every fixed $L \geq 2$ and fixed row $q \geq 2$,*

$$\frac{B_{L,X}(q)}{\sqrt{X}} = a_q \frac{(-1)^{L-1}}{(L-1)!} (\log \log X)^{L-1} + O_{L,q}((\log \log X)^{L-2}). \quad (22)$$

Consequently no fixed depth can simultaneously restore canonical parity and leave an eventually nonnegative current block:

- *if L is even, the recursive parity is canonical but $B_{L,X}(q) < 0$ for all sufficiently large X ;*
- *if L is odd, the leading current term is positive but the recursive packets retain odd parity.*

Proof. Insert (19) into (18). The principal part is

$$a_q \sqrt{X} \sum_{r=0}^{L-1} (-1)^r \sum_{\substack{m \leq X/q \\ \omega(m)=r}} \frac{1}{m}.$$

Lemma 6.3 shows that the term of highest degree in $H_{67}(X)$ is the $r = L - 1$ term. The accumulated error from the $O_q(\log(2X/m))$ remainder is, by (21), $o(\sqrt{X}(\log \log X)^{L-1})$. Since $H_{67}(X) = \log \log X + O(1)$, (22) follows. The parity statements come from Theorem 6.1. $\square$

The first repaired depth already fails at a moderate explicit endpoint.

Theorem 6.5 (Directed two-level counterexample). *At $X = 200000$, the depth-two current block satisfies*

$$B_{2,200000}(2) < -11, \quad B_{2,200000}(3) < -2. \quad (23)$$

Proof. The accompanying program evaluates (18) with 256-bit MPFR intervals and directed rounding for every square root, logarithm, addition, multiplication, division, and subtraction. It includes all 9574 active rough primes in row two and all 6627 in row three. The retained intervals are

$$\begin{aligned} B_{2,200000}(2) &\in [-11.2745354467343290, -11.2745354467343289], \\ B_{2,200000}(3) &\in [-2.1151635282980810, -2.1151635282980809]. \end{aligned}$$

The upper endpoints imply (23). $\square$

Firewall 6.6. *Advancing by a fixed even number of primes repairs the sign of the recursive packet but cannot yield a globally positive local current producer. Increasing the fixed depth merely postpones the same obstruction. A viable successor must balance parity globally, use a depth growing with the endpoint, or retain sign inside a non-scalar positive state.*

7 A clean global parity-aware gluing theorem

The fixed-depth no-go leaves a natural exact architecture: expand the finite source at a fixed endpoint, retain the cumulative parity of every atom, and solve one global common-source problem before any signed observation.

Let E and O be the actual positive even and odd source atoms after all rough-history swaps, with nonnegative source coefficients a_e and b_o . Give atom i a strictly positive target t_i , a score s_i , and a row vector r_i in a closed finite-dimensional cone K . In the conclusion-producing application one may take $K = \mathbb{R}_+^2$ for rows two and three, or $K = \mathbb{R}_+^{65}$ for rows 2 through 66.

A global exact-target removal is a coefficient vector

$$0 \leq u_e \leq a_e, \quad \sum_{e \in E} u_e t_e = \sum_{o \in O} b_o t_o. \quad (24)$$

Use the same u_e in target, score, and every row.

Theorem 7.1 (Global parity-aware common-source gluing). *Suppose a vector u satisfying (24) also satisfies*

$$\sum_e u_e r_e - \sum_o b_o r_o \in K, \quad \sum_e u_e s_e \leq \sum_o b_o s_o. \quad (25)$$

Then the signed source has the exact positive typed decomposition

$$(E - O) = (\nu; B; \sigma), \quad (26)$$

where

$$\nu_e = a_e - u_e \geq 0, \quad (27)$$

$$B = \sum_e u_e r_e - \sum_o b_o r_o \in K, \quad (28)$$

$$\sigma = \sum_o b_o s_o - \sum_e u_e s_e \geq 0. \quad (29)$$

Its row marginal is

$$R(E) - R(O) = R(\nu) + B \in K. \quad (30)$$

Conversely, if the compact feasible set defined by (24)–(25) is empty, finite-dimensional separation supplies a dual functional that excludes every common-source removal, not merely a chosen greedy transport.

Proof. Split each even coefficient once as $a_e = u_e + (a_e - u_e)$. Exact target equality uses all odd target and leaves the positive residual source ν . The two inequalities in (25) define the target-null positive row bonus B and score surplus σ , giving (30). For the converse, the coefficient box is compact and convex, the target condition is affine, and the cone constraints are closed and convex. If their intersection is empty, the finite-dimensional strong separation theorem gives the asserted dual certificate. $\square$

Definition 7.2 (Global parity-Hall producer). *The statement $\text{GPHT}_{2,3}$ asserts that the complete, parity-correct finite source at every endpoint admits the common-source certificate of Theorem 7.1 with $K = \mathbb{R}_+^2$ for rows two and three.*

Corollary 7.3 (Full-recursion reduction). *If $\text{GPHT}_{2,3}$ holds for every sufficiently large endpoint, then $c_X(2), c_X(3) \geq 0$ for every sufficiently large X .*

Proof. The full rough-prime tree is finite at fixed X and is a disjoint expansion of the original squarefree source. Activation and coefficient covariance are exact by Theorem 3.2. Apply Theorem 7.1 once after all history parities have been incorporated. Equation (30) gives the two row inequalities. $\square$

Remark 7.4. *The theorem is a genuine architectural repair but not a proof of $\text{GPHT}_{2,3}$. Unlike the predecessor, it does not commute a nonlinear Hall map through the rough tree and does not reuse a root base at separate leaves. The remaining producer is global and finite at each endpoint; it can emit either a common positive packing or an exact dual separator.*

8 The exact two-row Mellin–Landau consumer

This section is independent of the projective source architecture. It proves that eventual nonnegativity of rows two and three implies the Riemann Hypothesis.

8.1 Fixed-row Mellin transform

For fixed $j \geq 2$ and $\Re s > 1/2$, termwise integration is absolute and

$$\int_m^\infty \log \frac{X}{m} X^{-s-1} dX = \frac{m^{-s}}{s^2}. \quad (31)$$

With $z = s + 1/2$, define

$$H_j(z) = A_j j^{-z} - B_j(j+1)^{-z} + C_j \sum_{m \geq j+2} m^{-z}, \quad (32)$$

$$P_j(z) = A_j j^{-z} - B_j(j+1)^{-z} - C_j \sum_{m=1}^{j+1} m^{-z}. \quad (33)$$

Then $H_j(z) = C_j \zeta(z) + P_j(z)$.

Theorem 8.1 (Reciprocal-zeta row transform). *For $\Re s > 1/2$,*

$$\boxed{\int_1^\infty c_X(j) X^{-s-1} dX = \frac{C_j}{s^2} + \frac{P_j(s+1/2)}{s^2 \zeta(s+1/2)}}. \quad (34)$$

The right side gives meromorphic continuation to $\Re s > 0$ and is analytic at every positive real s .

Proof. Integrating (2) and using (31) gives $H_j(s+1/2)/s^2$. In (1), substitute $X = kY$ and sum the absolutely convergent Dirichlet series

$$\sum_{k \geq 1} \frac{\mu(k)}{k^{s+1/2}} = \frac{1}{\zeta(s+1/2)}.$$

Since $H_j = C_j \zeta + P_j$, equation (34) follows. On the positive real axis, $\zeta(s+1/2)$ has no zero; its pole at $s = 1/2$ becomes a zero of $1/\zeta$. $\square$

8.2 Two-row noncancellation

For rows two and three, let $x = 2^{-z}$ and $y = 3^{-z}$. Direct simplification gives

$$P_2(z) = 2x - 1 - y, \quad 3P_3(z) = 5y - x - 1 - 3x^2. \quad (35)$$

Theorem 8.2 (No common open-strip cancellation). *If $0 < \Re z < 1$, then $P_2(z)$ and $P_3(z)$ do not vanish simultaneously.*

Proof. If $P_2 = 0$, then $y = 2x - 1$. Substitution in the second equation of (35) gives

$$-3x^2 + 9x - 6 = -3(x-1)(x-2) = 0.$$

But $1/2 < |2^{-z}| < 1$ in the open strip, excluding both $x = 1$ and $x = 2$. $\square$

8.3 A self-contained Landau lemma

Lemma 8.3 (Landau for a nonnegative Laplace density). *Let $g : [0, \infty) \rightarrow [0, \infty)$ be locally integrable and let*

$$F(s) = \int_0^\infty g(t)e^{-st} dt$$

have finite abscissa of convergence σ_c . Then F cannot be holomorphic in a neighborhood of the real point σ_c .

Proof. Assume F is holomorphic in a disk of radius $R > 0$ centered at a real $\sigma_1 > \sigma_c$ with $\sigma_1 - \sigma_c < R$. Cauchy's estimate gives

$$\frac{|F^{(n)}(\sigma_1)|}{n!} \leq MR^{-n}$$

for some M . Positivity permits differentiation under the integral and removes absolute values:

$$(-1)^n F^{(n)}(\sigma_1) = \int_0^\infty t^n g(t) e^{-\sigma_1 t} dt \geq 0.$$

For $0 < \delta < R$, monotone convergence yields

$$\begin{aligned} \int_0^\infty g(t) e^{-(\sigma_1 - \delta)t} dt &= \sum_{n \geq 0} \frac{\delta^n}{n!} \int_0^\infty t^n g(t) e^{-\sigma_1 t} dt \\ &\leq M \sum_{n \geq 0} (\delta/R)^n < \infty. \end{aligned}$$

Choosing $\delta > \sigma_1 - \sigma_c$ contradicts the definition of σ_c . □

By the change of variables $X = e^t$, the same statement applies to nonnegative Mellin densities $f(X)X^{-s-1} dX$.

Theorem 8.4 (Two-row positivity implies RH). *If $c_X(2)$ and $c_X(3)$ are nonnegative for every sufficiently large X , then the Riemann Hypothesis holds.*

Proof. Removing a compact initial interval changes each Mellin transform by an entire function. Apply Lemma 8.3 to the nonnegative tail. Since the meromorphic formula (34) is analytic at every positive real point, the abscissa of convergence is at most zero, so the tail transform is holomorphic in $\Re s > 0$.

If ρ were a nontrivial zero with $\Re \rho > 1/2$, then $s_0 = \rho - 1/2$ would lie in $\Re s > 0$. At least one of $P_2(\rho)$ and $P_3(\rho)$ is nonzero, so the corresponding expression (34) has a pole at s_0 , a contradiction. The functional equation and conjugation symmetry then place every nontrivial zero on $\Re s = 1/2$. □

Corollary 8.5. *The global parity-Hall producer $\text{GPHT}_{2,3}$ implies the Riemann Hypothesis.*

9 What survives and what is withdrawn

Interface	Status
Canonical component-row formula	exact
Rough-history coefficient magnitude	exact
Rough-history row activation	exact
Cumulative parity $(-1)^{ h }$	exact
Canonical Target–Lorenz terminal map	one orientation only
Parity-blind arbitrary-history gluing	refuted as an argument
Full-row nonnegativity $c_X(j) \geq 0$	open, not refuted
Depth- L parity-aware source identity	exact
Fixed-depth local positive producer	impossible
Global producer GPHT _{2,3}	open and RH-bearing
Fixed-row reciprocal-zeta Mellin transform	exact
Rows two and three noncancellation	exact
Riemann Hypothesis	unproved

The directed Target–Lorenz tail calculations from the predecessor remain valid at their stated canonical-orientation scope. They do not repair the sign cocycle. Likewise, the exact coefficient contraction and one-use source ledgers remain useful. The failed step is the claim that one canonical terminal map may be substituted after arbitrary rough histories.

10 Deterministic verification

The accompanying verifier uses Python’s arbitrary-precision integers and `Fraction` class for the dyadic target obstruction, together with direct 256-bit MPFR intervals for the fixed-depth current block. It checks:

- (1) 16 ordered rough histories over the fixture primes 67, 71, 73, 79;
- (2) 309120 exact coefficient, ratio, activation, and parity identities;
- (3) all 239 active P_{61} atoms in the leaf (71, 13);
- (4) the rational interval inequality (14);
- (5) a disjoint and exhaustive two-level partition of every subset of the four-prime fixture;
- (6) a 256-bit directed MPFR evaluation of the complete depth-two current block at $X = 200000$;
- (7) the exact polynomial factorization in (35);
- (8) hostile mutations that drop history parity, reverse Hall capacity, or terminalize a one-prime leaf parity-blindly.

The retained verdict is

```
PASS_X_96500_PARITY_COVARIANT_GLUE_AUDIT
918b3ccfb7f99d764e01302ffc2f358fda0e3e2ccfc27eeb038799cf0ee05293
E_T(71,13)-O_T(71,13)>17
```

The finite fixture authenticates the exact formulas and the counterexample. It does not prove GPHT_{2,3}, two-row positivity, or RH.

11 Conclusion

The hostile reconstruction resolves the parity and activation question sharply. Row activation and coefficient magnitude are preserved under every rough-prime placement. The nontrivial interface is the sign cocycle. A single preceding rough prime reverses the canonical one-prime leaf, and the published Target–Lorenz map cannot be reversed even at target-mass level for the explicit leaf (71, 13).

The obvious fixed-depth repair is not viable. Odd depth leaves the recursive parity reversed; even depth restores it but creates an eventually negative current Euler truncation. The exact row-two and row-three counterexample at $X = 200000$ makes this obstruction concrete. Thus the route cannot be repaired by merely grouping two, four, or any other fixed number of rough levels.

The clean surviving architecture is global. At a fixed endpoint the complete source tree is finite, so every cumulative parity can be incorporated before one common-source transport is applied. Theorem 7.1 carries all source, target, score, and row coordinates with one coefficient vector and has an exact dual alternative. Its two-row producer $\text{GPHT}_{2,3}$ is the remaining arithmetic theorem. The downstream analytic chain is complete: $\text{GPHT}_{2,3}$ implies eventual positivity of rows two and three, and that positivity implies RH through the reciprocal-zeta Mellin transforms and Landau’s theorem. No proof of $\text{GPHT}_{2,3}$ is contained here. Accordingly, RH remains unproved.

A Exact interval certificate schema

For completeness, the target certificate is determined by the following finite algorithm.

- (i) Generate every squarefree divisor d of P_{61} with $d \leq 923$.
- (ii) Determine $\mu(d)$ by parity of its prime subset.
- (iii) Enclose each square root in (11) by the exact dyadic method in Theorem 5.1.
- (iv) Apply outward rational interval arithmetic to obtain t_d .
- (v) Sum t_d separately over $\mu(d) = 1$ and $\mu(d) = -1$.
- (vi) Subtract the upper odd endpoint from the lower even endpoint.

Every operation is an integer or rational operation. The certificate is therefore independent of the host floating-point environment.

B Repository provenance and claim boundary

The reconstruction froze the predecessor projective full-row proposal at head 20646a78c3e8843001cb49ea0c9741f6d0d446f7, stacked on the projective native-Lorenz proposal at ef18bdda5a65334695008e4c1f5986833160d84f. The observed main branch at the audit cutoff was 994bd4bedcc8b61cebabaf005cc69225fc3fe459.

The present manuscript makes no claim that the full native row is negative. It withdraws only the parity-blind proof. Its new theorem labels are:

L-96500	parity-covariant projective gluing
L-96501	activation and coefficient covariance
R-96500	explicit odd-history obstruction
L-96502	exact depth- L parity expansion
R-96501	fixed-depth positive-reset no-go
L-96503	global parity-aware common-source gluing
T-96500	GPHT _{2,3} $\Rightarrow$ two-row positivity $\Rightarrow$ RH

References

- [1] B. Riemann, *Ueber die Anzahl der Primzahlen unter einer gegebenen Groesse*, Monatsberichte der Berliner Akademie, 1859.
- [2] E. Landau, classical singularity theorem for Dirichlet and Laplace transforms with nonnegative coefficients. A self-contained form sufficient here is proved as Lemma 8.3.